

Recalibrating Tail Risk Forecasts under Temporal Dependence

Conformal Audit of Foundation Models for VaR Forecasting

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The Thinker of Hamangia: Does the Model Know Its Own Limits?

The Oldest Known Act of Self-Examination

- The Thinker & The Sitting Woman, Hamangia culture, Cernavodă, Romania, ca. 5000 BC
- $\hat{q}_V$ asks the same question of forecasters: *do you know your own limits?*

From Philosophy to Operations

- Self-examination is not only philosophical — in risk forecasting it becomes operational
- A model must know whether its own 1% tail is calibrated before its VaR can be trusted



Why Tail Calibration Matters

The 1% Tail Is Where Capital Lives

- The 1% Value-at-Risk (VaR) determines:
  - Basel III capital charges ($m_c \times \widehat{\mathrm{VaR}}$)
  - Margin requirements for centrally cleared derivatives
  - Stress test thresholds
- A forecaster that is accurate on average can still be catastrophically wrong at the 1% tail

The Problem Is Structural

- Foundation models: pre-trained on non-financial data that under-represents fat tails
- Parametric models: conditional variance dynamics do not guarantee tail shape
- Miscalibration at the 1% level has direct financial consequences

Miscalibration Is Expensive

Basel Traffic Light: Violations ⇒ Capital Multiplier m_c

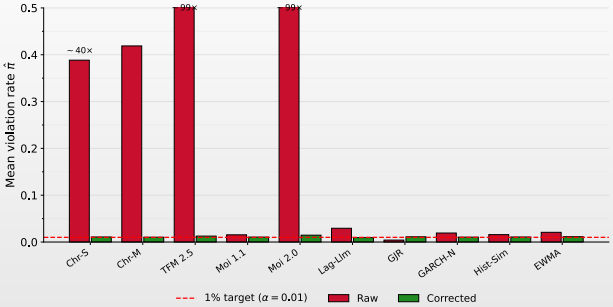
- **Green** ($\leq$ 4 viol./250d): $m_c = 3.00$, baseline
- **Yellow** (5–9 viol.): $m_c \in [3.40, 3.85]$, penalty zone
- **Red** ($\geq$ 10 viol.): $m_c = 4.00$, internal model disallowed

One Zone Downgrade Costs 17.8% of Capital

$$\text{Capital charge}_t = m_c \, \widehat{\text{VaR}}_t$$

- S&P 500, 2018–2026: Yellow → Green saves **17.8%** on cumulative reserves, no model change
- Red zone is worse: internal model disallowed, fallback to standardised approach

Raw Violation Rates: The Calibration Failure



Where Forecasters Fail (Raw, Before Any Correction)

- Four forecasters miss severely: Chronos-Small, Chronos-Mini, TimesFM 2.5, Moirai 2.0
- Violation rates span **0.4%–99%**: two orders of magnitude
- Moirai 1.1: $\hat{\pi} = 1.5\%$—output interface matters
- GJR-GARCH: over-conservative ($\hat{\pi} = 0.4\%$), only model below target

Tail Sparsity: Why 1% VaR Is Hard

The Binding Constraint Is αT, Not T

- $\alpha = 0.01$, $T \approx 1{,}500$ test days $\Rightarrow$ only ~ 15 expected tail observations
- More history does not produce proportionally more tail signal
- All tail corrections—parametric or not—operate on these ~ 15 observations

The Obvious Fixes Do Not Work

- **Retrain?** Foundation models are black-box; no tail-aware retraining recipe exists
- **More data?** Binding constraint is αT; doubling T adds ~ 15 more tail observations
- **Richer models?** Any correction inherits a variance penalty growing with parameter count

Effective Tail Sample Size

Heuristic: Bias–Variance at the Tail

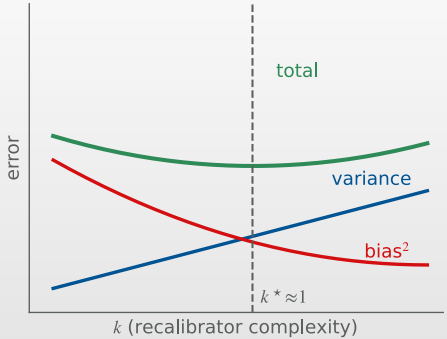
- Variance of a k-parameter tail correction scales as

$$\mathrm{Var} \sim \frac{k}{\alpha T}$$

- At $\alpha T \approx 15$: each extra parameter adds $\sim$7% variance

Implication

- Under tail sparsity, low-dimensional corrections are statistically favoured
- Tail sparsity strongly favours k=1
  - Additional parameters add variance faster than they reduce bias



Variance dominates at $\alpha T \approx 15$;
k=1 minimises estimated total error (heuristic).

Research Question

The Question

- How much tail information does a forecaster actually contain?
- A one-parameter conformal shift, sized against raw VaR, partitions forecasters into two regimes
- The shift magnitude measures tail information content: it is an audit statistic, not just a corrector
- The conformal recalibration step is forecast-agnostic; asymptotic validity requires β-mixing of the score process (Theorem 3.3)

Our Answer: A Single Scalar $\hat{q}_V$

- **Recalibrates** VaR forecasts without retraining
- **Audits** whether the base model contains usable tail signal
- **Separates** signal-preserving models from replacement-regime models

γνῶθι σαυτόν (Know Thyself, Delphi) — $\hat{q}_V$ is the forecaster's audit.

One Scalar Correction: $\hat{q}_V$

The Correction

$$\widehat{\mathrm{VaR}}_t^{cp} = -\left(\hat{q}_t^{lo} - \hat{q}_V^{\mathrm{stat}}\right)$$

- One parameter, one shift, applied to any base forecaster's lower quantile $\hat{q}_t^{lo}$
- Forecaster-agnostic: same formula for GARCH, EWMA, Chronos, TimesFM, Moirai, Lag-Llama

Sign Tells the Direction; Magnitude the Severity

- $\hat{q}_V^{\mathrm{stat}} > 0$: under-covers $\rightarrow$ **widen** VaR (e.g. Chronos-Small: $+0.038$)
- $\hat{q}_V^{\mathrm{stat}} < 0$: over-conservative $\rightarrow$ **tighten** VaR (e.g. GJR-GARCH: -0.007)
- $\hat{q}_V^{\mathrm{stat}} \approx 0$: near-calibrated (e.g. Moirai 1.1: $+0.004$)

Dual Role

- $\hat{q}_V$ is simultaneously a **correction parameter** and a **diagnostic statistic**—signed, continuous, in return units

Static vs Rolling Conformal: Audit vs Deployment

Two Uses of the Same Object

- **Static** $\hat{q}_V^{\text{stat}}$: cross-sectional audit of model tail failure.
  - Fixed calibration window (70/30 split); ranks models, identifies replacement regime
- **Rolling** $\hat{q}_{V,t}^{\text{roll}}$: operational recalibration under changing regimes.
  - 250-day trailing window, re-estimated daily; adapts to local score distribution shifts

Static Conformal

$$\hat{q}_V^{\text{stat}} = Q_{1-\alpha}\left(\{s_t^V\}_{t=1}^{n}\right), \qquad \widehat{\text{VaR}}_t^{cp} = -\left(\hat{q}_t^{lo} - \hat{q}_V^{\text{stat}}\right)$$

Rolling Conformal ($w = 250$ days)

$$\hat{q}_{V,t}^{\text{roll}} = Q_{1-\alpha}\left(\{s_\tau^V\}_{\tau=t-w}^{t-1}\right), \qquad \widehat{\text{VaR}}_t^{roll} = -\left(\hat{q}_t^{lo} - \hat{q}_{V,t}^{\text{roll}}\right)$$

Data Universe

Asset Universe (Selected for Tail-Risk Diversity)

Class	#	Examples
Equity	11	SP500, STOXX, DAX, Nikkei
Bond	3	TLT, CBU0.L, IBGL
Commodity	4	Gold, WTI, NatGas, DJCI
Crypto	2	BTC, ETH
FX	4	EUR/USD, GBP/USD, USD/JPY, AUD/USD

Why This Mix

- Thin-tailed (FX, bonds) to fat-tailed (crypto, NatGas)

Scale

- 10 forecasters $\times$ 24 assets $=$ **240 pairs**
- $\approx$ 36,600 pooled test days per TSFM

Configuration

- Context: 512 days (TSFMs)
- Calibration / test split: 70% / 30% (chronological)



Ten Forecasters, One Conformal Wrapper

Time Series Foundation Models (zero-shot)

- **Sampling-based** (1,000 MC draws): Chronos-S/M, Moirai 1.1, Lag-Llama (Student-t head)
- **Quantile grid** (+ Student-t tail completion): TimesFM 2.5, Moirai 2.0

Classical Benchmarks (250-day rolling)

- GJR-GARCH (skew-t innovations)
- GARCH-N and EWMA
- Historical Simulation

Key Distinction

- TSFMs produce quantiles without per-asset estimation: scalable but untested at the tail
- Classical models fit per-asset parameters: well-tested but not scalable across large universes
- The conformal wrapper is applied identically to all ten forecasters

Experimental Protocol

Static (Audit) Protocol

- 70% calibration / 30% test (chronological)
- $\approx$ 1,500 test days per asset
- $\hat{q}_V^{\text{stat}}$ estimated once on calibration set

Rolling (Operational) Protocol

- 250-day trailing window, re-estimated daily
- $\hat{q}_{V,t}^{\text{roll}}$ adapts to local score distribution

Reproducibility

- HuggingFace checkpoints pinned; no fine-tuning
- All checkpoints predate evaluation period
- Yahoo Finance data; full pipeline in Quantlet repo

Evaluation

- Kupiec, Christoffersen CC, Basel Traffic Light
- Quantile Score (pinball loss)
- Basel Green zone rate across 240 pairs

Conformal Prediction: Mechanics

Three Steps

1. **Score**: $s_t^V = \hat{q}_t^{lo} - r_t$ on calibration set

   Positive when the return falls below the predicted quantile (violation)
2. **Sort**: $s_{(1)}^V \leq \ldots \leq s_{(n)}^V$
3. **Apply**: $\hat{q}_V^{\text{stat}} = \lceil (1-\alpha)(n+1) \rceil$-th order statistic

$$\widehat{\text{VaR}}_t^{cp} = -(\hat{q}_t^{lo} - \hat{q}_V^{\text{stat}})$$

Intuition

- Black-box: no model access required
- $\hat{q}_V^{\text{stat}}$ absorbs systematic bias
- If raw VaR missed by 0.8 units on average, $\hat{q}_V^{\text{stat}} \approx 0.8$ widens VaR by exactly that amount

(numerical example in backup)

One-Sided Conformal: Why It Matters for VaR

Why One-Sided?

- VaR is a lower-tail problem: only downside exceedances count
- Two-sided CQR (Romano et al. 2019) splits the coverage budget across both tails
- One-sided design targets the regulatory loss-side asymmetry

The Score $s_t^V = \hat{q}_t^{lo} - r_t$

- $s_t^V > 0$: violation day (return fell below predicted quantile)
- $s_t^V \leq 0$: no violation (VaR held)
- $\hat{q}_V^{\text{stat}}$ is the $(1-\alpha)$-quantile of these scores: captures how far the VaR misses

One Parameter, Full Diagnostic

- Magnitude: $+0.038$ means "widen by 3.8 percentage points of daily return"
- In return units: maps directly to capital allocation

Temporal Dependence: Validity without Exchangeability

The Practical Point

- We do not rely on exchangeability — under weak temporal dependence (β-mixing), the conformal quantile remains asymptotically valid up to a dependence penalty:

$$\mathbb{P}\left(r_{n+1} \geq -\widehat{\mathrm{VaR}}^{cp}_{n+1}\right) \geq 1 - \alpha - \Delta_n, \qquad \Delta_n \leq C\sqrt{\tfrac{\log n}{n}} + \beta(g_n)$$

- The correction is estimated from recent forecast errors, not from an i.i.d. idealisation
- Stationary GARCH satisfies the mixing condition (Lindner 2009)

Empirical Validation

- Theoretical floor at $n_{\mathrm{cal}} \approx 4{,}300$: $\approx 88\%$
- Empirical coverage is close to the 99% target, with deviations concentrated in unstable replacement-regime models

(β-mixing details) | (full theorem) | (proof sketch)

$\hat{q}_V$ as Diagnostic: More Than a Correction

$\hat{q}_V^{\text{stat}}$ Is Not a Test Statistic

- **Signed**: direction of miscalibration (> 0: widen, < 0: tighten)
- **Continuous**: detects miscalibration before zone-crossing thresholds
- **In return units**: maps directly to capital allocation
- No p-value or critical value needed; physically interpretable

What $\hat{q}_V^{\text{stat}}$ Reveals

- $\hat{q}_V^{\text{stat}} \approx 0$ (Moirai 1.1, Hist. Sim.): base forecast nearly calibrated at the tail
- $\hat{q}_V^{\text{stat}} \gg 0$ (TimesFM 2.5, Moirai 2.0): base forecast contains no usable tail signal
- $\hat{q}_V^{\text{stat}} < 0$ (GJR-GARCH): over-conservative, correction tightens

Kupiec Diagnoses; $\hat{q}_V$ Prescribes

- Kupiec tells you *whether* VaR fails; $\hat{q}_V$ tells you *by how much* and *in which direction*—and fixes it

$\hat{q}_V^{\text{stat}}$ Ranking and Cluster Structure

Model	$\overline{\hat{q}_V^{\text{stat}}}$	Cluster
TimesFM 2.5	$+0.102$	replacement
Moirai 2.0	$+0.093$	replacement
Chronos-Mini	$+0.039$	replacement
Chronos-Small	$+0.038$	replacement
Lag-Llama	$+0.011$	signal-preserving
EWMA	$+0.006$	signal-preserving
GARCH-N	$+0.006$	signal-preserving
Moirai 1.1	$+0.004$	signal-preserving
Hist. Sim.	$+0.004$	signal-preserving
GJR-GARCH	-0.007	outlier (over-conserv.)

Two Clusters, One Outlier

- 9/10 under-cover ($\hat{q}_V^{\text{stat}} > 0$); only GJR-GARCH over-conservative
- Magnitude spans two orders: 0.004 (Moirai 1.1) to 0.102 (TimesFM 2.5)
- Clear gap (0.012–0.038) separates signal-preserving from replacement regime

$\hat{q}_V^{\text{stat}}$ as Diagnostic: Partial $R^2 = 53.4\%$

Regression Model (panel of 240 forecaster–asset pairs)

$$\Delta\text{QS}_i = \beta_0 + \beta_1 \hat{q}_{V,i}^{\text{stat}} + \beta_2 \hat{\pi}_{\text{raw},i} + \varepsilon_i$$

- $\Delta\text{QS}_i > 0$: conformal correction improved quantile score (pinball loss)
- $\hat{\pi}_{\text{raw},i}$: raw violation rate (Kupiec), the control variable
- Clustered SEs (by asset and by model, reported separately)

The Headline Result

- Overall $R^2 = 78.2\%$; **Partial R^2 of $\hat{q}_V^{\text{stat}}$ = 53.4%**
- $\hat{q}_V^{\text{stat}}$ explains most of ΔQS even after controlling for raw violation rates
- $\hat{q}_V^{\text{stat}}$ is a *genuine diagnostic*, not just a recalibration parameter

Why It Beats Violation Counts

- Kupiec's $\hat{\pi}_{\text{raw}}$ is binary (pass/fail) and unsigned; $\hat{q}_V^{\text{stat}}$ is continuous, signed, and in return units

Two Regimes: Signal-Preserving vs Replacement

Replacement Ratio $R = |\hat{q}_V^{\text{stat}}| / |\widehat{\text{VaR}}^{\text{raw}}|$

- $R < 1$ (**signal-preserving**): correction small relative to raw VaR; base forecast retains content
- $R \gg 1$ (**replacement**): correction dominates; conformal quantile overrides the base

Empirical Classification

- **Signal-preserving**: Moirai 1.1, Lag-Llama, GJR-GARCH, GARCH-N, Hist. Sim., EWMA ($\bar{R} = 0.11$–0.36)
- **Replacement**: Chronos-S/M, TimesFM 2.5, Moirai 2.0 ($\bar{R} = 3.2$–23.5)

Headline

- Signal-preserving: 127/144 Green (88.2%); CC pass 86.1%
- Replacement: 76/96 Green (79.2%); CC pass 40.6%

(full table) | (operational) | (sensitivity)



Basel Traffic Light: Raw vs Corrected

Each cell: annualized violation count per model–asset pair, $\alpha = 0.01$



The Transformation

- **Raw**: 63/240 Green (26.2%)—most TSFMs in Red zone
- **Static conformal**: 203/240 Green (**84.6%**)
- **Rolling conformal**: 234/240 Green (**97.5%**)

Static Conformal: Cross-Sectional Audit

Per-Asset-Class Breakdown (Static 70/30 Split)

Asset class	N	Corr. $\hat{\pi}$	Green
Equity	11	.010	87/99 (88%)
Bond	3	.010	23/27 (85%)
Commodity	4	.016	23/36 (64%)
Crypto	2	.005	18/18 (100%)
FX	4	.009	36/36 (100%)
All	24	.011	**203/240 (84.6%)**

Limitation: Static Does Not Adapt

- Commodities weakest (64%): NATGAS and WTI regime-switching volatility
- Static $\hat{q}_V^{\text{stat}}$ is a fixed shift; cannot track changing score distributions
- Static conformal is primarily an **audit tool**; operational deployment needs rolling



Rolling Conformal: Operational Deployment

Headline: 234/240 Green (97.5%)

- Chronos-S/M: 24/24 Green (from 0/24 raw)
- Commodities: 64% $\rightarrow$ **94%** (the weakest static class is nearly resolved)
- Bonds: 85% $\rightarrow$ **100%**

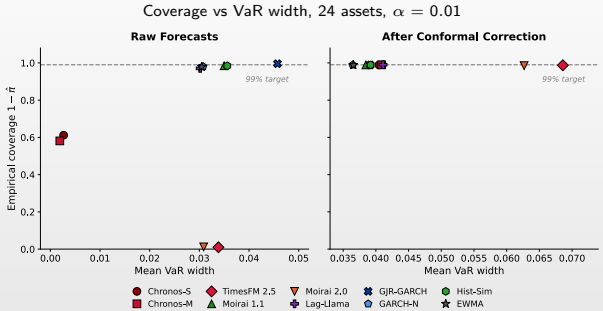
Why Rolling Works

- $\hat{q}_{V,t}^{\text{roll}}$ rises during volatility spikes, falls in calm periods
- Operates on forecast errors, not raw returns: corrects for model bias, not market state
- Robust to window choice: $w \in \{125, 250, 500\}$ gives Green 97–99%

Coverage Bounds (finite-sample caveat)

- **Time-average**: $\frac{1}{T}\sum_t \mathbf{1}\{\text{viol.}\} \rightarrow \alpha$ at rate $O(T^{-1/2})$
  - Via OGD regret; no distributional assumptions
- **Per-window**: drift diagnostic $\hat{\delta}_w(t)$ identifies when the bound weakens

Calibration–Efficiency Frontier



Coverage Convergence at Different Width Costs

- **Raw**: Chronos at ∼60%, TimesFM 2.5 & Moirai 2.0 at ∼1%
- **Corrected**: all 10 forecasters converge to 99% coverage
- **Replacement regime**: 99% achieved at ∼1.7× wider intervals
- **GJR-GARCH** (fitted per asset): only model where correction *tightens* the interval

Within-Family Contrast: Moirai 1.1 vs Moirai 2.0

	Moirai 1.1	**Moirai 2.0**
Predictive interface	1,000 MC samples	Nine-decile quantile grid
Raw violation rate $\hat{\pi}$	1.5%	98.8%
$\hat{q}_V^{\text{stat}}$ (median across assets)	$+0.004$	$+0.093$
$\bar{R}$ / classification	0.11 / signal-preserving	3.2 / replacement
CC pass (Christoffersen)	13/24	0/24

Takeaway

- Architecture and training data approximately held fixed across versions
- The predictive interface—sample-based vs quantile-grid—emerges as a dominant driver of tail miscalibration
- Same foundation model, opposite regime classification

Baselines: All Comparators

Method	Complexity	$\hat{\pi}$	Kup. rej.	QS ($\times 10^4$)	Width	Green %
Raw (no correction)	—	.289	199/240	74.99	.028	26.2
Conformal static (ours)	**1 shift**	**.011**	**100/240**	**6.07**	.044	**84.6**
Conformal rolling w=250	**1 shift**	**.010**	**13/240**	5.63	.044	**97.5**
ACI (Gibbs–Candès 2021)	adaptive	.010	6/216	5.37	.040	96.3
GAMLSS-SST	7 params	.010	—	5.38	—	88.0
Scale (multiplicative)	1 scale	.264	215/216	28.51	.023	11.6
Historical quantile	0 params	.010	99/216	5.89	.041	81.0
QR on residuals	2 params	.012	68/216	5.29	.038	81.5
GBM-QR (Gradient Boosting)	nonlinear	.018	128/216	5.48	.033	48.6
EVT-POT (per-asset)	param.	.014	10/24	5.5	.036	75.0
Filtered HS (per-asset)	param.	.016	13/24	5.6	.036	54.2

Pattern

- Green rates are **not monotone in complexity**: GBM-QR (nonlinear) = 48.6%; conformal (1 shift) = 97.5%
- ACI is competitive (96.3% Green); rolling conformal is more directly aligned with one-sided VaR exceedance control
- GAMLSS-SST matches $\hat{\pi}$ but lower Green (88% vs 97.5%) at higher complexity (7 vs 1 parameters)

Per-asset baselines evaluated on 24 assets only; panel methods on 240 model–asset pairs. (ACI detail) | (dedicated VaR detail)

GAMLSS-SST: Strong Baseline, Higher Complexity

GAMLSS-SST (Rigby–Stasinopoulos 2005)

- Skew Student-t with dependencies on μ, σ, ν, τ; 7 parameters
- Empirical: $\hat{\pi} = .010$, QS 5.38×10^{-4}, **88.0%** Green

Rolling Conformal

- 1 parameter; targets only the tail quantile
- $\hat{\pi} = .010$, QS 5.63×10^{-4}, **97.5%** Green

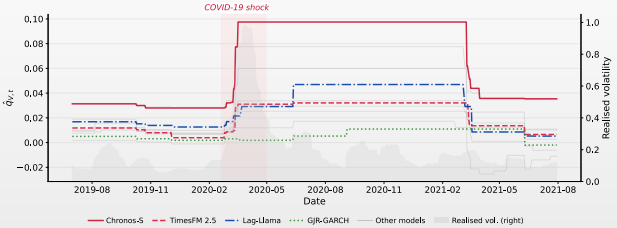
Assessment

- Rolling conformal offers **stable Basel-zone outcomes** (97.5% vs 88.0% Green) with lower complexity (1 vs 7 parameters)

Lesson

- At the 1% tail with $\alpha T \approx 15$ effective observations, temporal adaptation beats distributional complexity

COVID-19 Stress Test: The Trailing-Window Lag



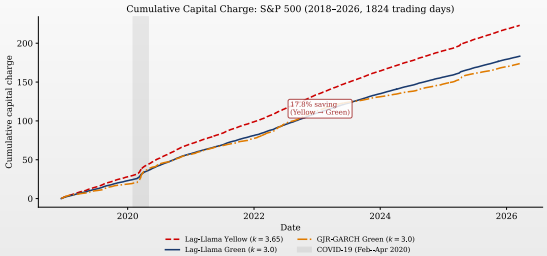
The Lag

- **Shock**: vol peaks at 0.98 annualised (8× pre-shock)
- 6/10 models peak within the acute window
- First $\approx w/2$ days under-protected: unavoidable

The Safeguard

- Drift diagnostic $\hat{\delta}_w(t)$ identifies the regime in real time
- **Operational rule**: shorten w or pause when $\hat{\delta}_w > 0.10$
- Post-shock: $\hat{q}_V^{\text{roll}}$ reverts once stress exits the window

Capital Impact: 17.8% Saving from Zone Reclassification



S&P 500, 1,824 trading days (2018–2026) ( )

- Cumulative charge: $\sum_t m_c \, \widehat{\mathrm{VaR}}_t$ where m_c is the Basel multiplier
- Red dashed: Lag-Llama Yellow zone ($m_c = 3.65$)
- Blue solid: Lag-Llama Green after conformal ($m_c = 3.00$)
- Orange dash-dot: GJR-GARCH Green ($m_c = 3.00$), tightest width
- Saving via multiplier channel: $(3.65 - 3.00)/3.65 = \mathbf{17.8\%}$; positive whenever Width/GJR < 1.22

Practitioner Guidance

Scenario 1: Single-Asset Regulatory VaR

- GJR-GARCH (fitted per asset) + single-split conformal: lowest width, strongest per-asset benchmark

Scenario 2: Large Universe, Zero-Shot

- Moirai 1.1 or Lag-Llama + rolling conformal; no per-asset estimation
- Screen with $R = |\hat{q}_V^{\text{stat}}|/|\widehat{\text{VaR}}^{raw}|$ before deployment

Scenario 3: Model Selection / Ensemble Screening

- $R < 1$: signal-preserving $R \gg 1$: effective replacement

Minimum Requirements

- $T \geq 1{,}000$, $f_c \geq 0.60$; monitor $\hat{\delta}_w(t)$, pause when >0.10

Limitations and Open Questions

Where Rolling Conformal Fails

- **Unprecedented regimes**: first $\approx w/2$ days under-protected (COVID lag ≈ 125 days)
- **Location only**: $T(q) = q + c$; no distributional reshaping
- **Conditional coverage not restored**
  - Marginal coverage converges, but violation clustering persists
  - Replacement-regime models pass CC on 0–5/24 assets

Formal Guarantees

- Rolling conformal lacks the finite-sample exchangeability guarantee of static split-conformal
- Two operational bounds substitute
  - OGD time-average $O(T^{-1/2})$
  - Per-window drift diagnostic $\hat{\delta}_w(t)$
- Weaker than finite-sample certificates; bound average miscoverage, not worst-case per-step

ES Extension

- VaR coverage is formally guaranteed; ES correction is heuristic
- Formal ES conformal guarantee remains an open problem (ES backtest)

Conclusion

Main Finding

- At the 1% tail, a one-parameter correction is not a simplification; it is the statistically appropriate complexity level

What $\hat{q}_V$ Adds

- Recalibrates VaR forecasts without retraining
- Audits whether the base model contains usable tail signal
- Separates signal-preserving models from replacement-regime models

Caveats

- Marginal coverage restored; **conditional coverage is not**—violation clustering persists
- GJR-GARCH (fitted per asset) retains a QS advantage; zero-shot TSFMs trade accuracy for scale
- Rolling bounds are time-average, not finite-sample certificates

Practical Message

- Use foundation models for scale, but audit their tails before trusting their risk forecasts

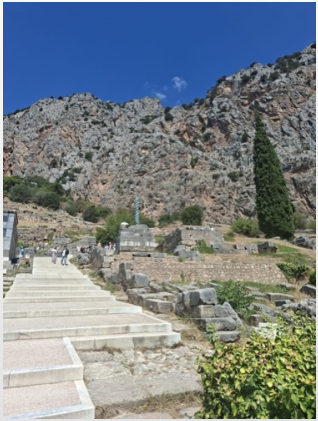
Who Calibrates the Oracle?

Three Layers

- **The forecast** — the oracle's words / the TSFM's output
- **The interpretation** — priests of Apollo / the predictive interface
- **The audit** — history's verdict / $\hat{q}_V$'s magnitude

$\gamma\nu\tilde{\omega}\theta\iota\ \sigma\alpha\upsilon\tau\acute{o}\nu$ — Know Thyself

- Modern forecasters lack the third layer
- $\hat{q}_V$ supplies it: continuous, signed, in return units



The Oracle of Delphi

Backup Slides

What Is β-Mixing?

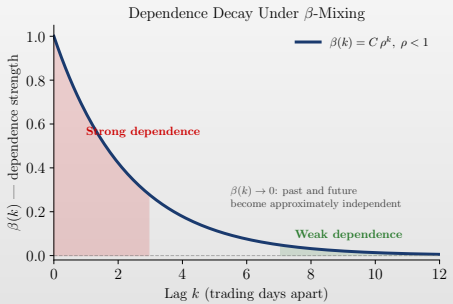
Notation

- $\{X_t\}_{t\in\mathbb{Z}}$: stationary time series
- $\mathcal{F}_{\le t}$: past up to t
- $\mathcal{F}_{\ge t+k}$: future from $t+k$ on

β-mixing coefficient at lag k

$$\beta(k) = \sup_t \mathbb{E}\left[\sup_{B\in\mathcal{F}_{\ge t+k}} \left|P(B \mid \mathcal{F}_{\le t}) - P(B)\right|\right]$$

- $\beta(k) \to 0$: distant observations $\approx$ independent
- **Geometric**: $\beta(k) \le C\rho^k$, $\rho < 1$; stationary GARCH satisfies this (Lindner 2009)



→ coverage guarantee

Coverage Guarantee under β-Mixing Dependence

Assumptions on the Score $S_t = \hat{q}_t^{lo} - r_t$

- **(A1)** Stationarity, geometric β-mixing: $\beta(k) \le C\rho^k$
- **(A2)** Score density bounded away from zero near $q_{1-\alpha}$
- **(A3)** Gap separation: g_n obs. between calibration and test

Theorem 1

Under (A1)–(A3), define $\widehat{\mathrm{VaR}}^{cp}_{n+g_n+1} = -(\hat{q}^{lo}_{n+g_n+1} - \hat{q}^{\mathrm{stat}}_V)$. Then

$$\mathbb{P}\left(r_{n+g_n+1} \ge -\widehat{\mathrm{VaR}}^{cp}_{n+g_n+1}\right) \ge 1 - \alpha - \Delta_n, \qquad \Delta_n \le C\sqrt{\frac{\log n}{n}} + \beta(g_n)$$

Empirical Validation

- Empirical coverage is close to the 99% target, with deviations concentrated in unstable replacement-regime models
- Stationary GARCH(1,1) satisfies the β-mixing condition (proof sketch) | (rolling guarantee)

Coverage Guarantee: Three-Step Proof Sketch

Assumptions: (A1) long-range dependence, (A2) quantile well-defined, (A3) theoretical gap ($g_n = 0$ in practice)

Step 1: Reduction to score control

- $S_t = \hat{q}_t^{lo} - r_t$; coverage $\Leftrightarrow \{S_{n+g_n+1} \leq \hat{q}_V^{\text{stat}}\}$

Step 2: Empirical $\rightarrow$ population quantile

- Bahadur under A1, A2: $|\hat{q}_V^{\text{stat}} - q_{1-\alpha}| = O_p\left(\frac{1}{f_S(q_{1-\alpha})}\sqrt{\log n/n}\right)$; f_S: score density; ρ_n: mixing remainder

Step 3: Combine and take expectation

- $\mathbb{P}(S_{n+g_n+1} \leq \hat{q}_V^{\text{stat}}) \geq 1 - \alpha - \Delta_n, \quad \Delta_n \leq C\sqrt{\log n/n} + \beta(g_n)$

Bound is conservative

- At $n_{\text{cal}} \approx 4{,}300$: theoretical floor $\approx 88\%$; empirical 98.9%

Corollary: GARCH(1,1) Satisfies the Assumptions

GARCH(1,1) Model and Notation

- $r_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$
  - α_1 — ARCH (shock sensitivity), β_1 — GARCH (persistence), $\varepsilon_t \overset{\text{iid}}{\sim} (0, 1)$

Stationarity $\Rightarrow$ β-Mixing

- If $\mathbb{E}[\log(\alpha_1 \varepsilon_t^2 + \beta_1)] < 0$, then $\{r_t\}$ is
  - Strictly stationary (Francq & Zakoïan 2019, Thm 2.10)
  - Geometrically β-mixing: $\beta_r(k) \leq a\rho^k$ (Lindner 2009)

Score Process Inherits Mixing

- $S_t = \hat{q}_t^{lo} - r_t$ is a measurable function of $\{r_s\}_{s \leq t}$
  - $\beta_S(k) \leq \beta_r(k-1)$: score mixing bounded by return mixing
- **Coverage guarantee applies with** $\Delta_n \leq C\sqrt{\log n / n}$

Application to Our Benchmarks

- **GJR-GARCH**: asymmetric threshold preserves stationarity under analogous top-Lyapunov condition; corollary applies
- **GARCH-N**: special case of GARCH(1,1) with Gaussian innovations; corollary applies directly
- **EWMA**: non-stationary ($\alpha_1 + \beta_1 = 1$); the static corollary does not apply, but the rolling variant remains operational

Extension: FIGARCH and Hyperbolic Mixing

When Geometric Mixing Fails

- The GARCH corollary requires geometric β-mixing: $\beta(k) \leq C\rho^k$
- **FIGARCH** (Fractionally Integrated GARCH): hyperbolic decay $\beta_r(k) = O(k^{-\gamma})$
- Long memory in volatility violates geometric mixing

Coverage Guarantee Still Applies

- The guarantee requires only $\rho_n \to 0$ (mixing remainder vanishes)
- Under hyperbolic mixing: $\Delta_n = O(n^{-\delta})$ for some $\delta > 0$
  - Slower than $O(\sqrt{\log n / n})$ but still valid
- Coverage guarantee holds; convergence rate degrades with fractional parameter

Practical Reading

- Long-memory volatility (commodities, FX): larger n_{cal} recommended
- Rolling variant provides additional adaptation

Δ_n Empirical Validation

Model	Asset	n_{cal}	$\hat{\rho}$	$\hat{\Delta}_n$	Guaranteed	Empirical
Chronos-S	S&P 500	4,254	0.00	0.111	87.9%	98.7%
Chronos-S	BTC	2,581	0.18	0.138	85.2%	99.9%
TimesFM 2.5	S&P 500	4,254	0.62	0.111	87.9%	98.9%
Moirai 1.1	DAX	4,299	0.23	0.110	88.0%	98.7%
Lag-Llama	FTSE	4,275	0.33	0.111	87.9%	98.9%
GJR-GARCH	S&P 500	4,438	0.67	0.109	88.1%	98.6%



Reading

- $\hat{\Delta}_n \approx 0.11$ across all pairs, essentially independent of $\hat{\rho}$
- Theoretical floor $\approx 88\%$; empirical coverage $> 98\%$
- Bound is directionally correct but conservative; the price of distribution-freeness

Practitioner Guide: Coverage Bounds

n_{cal}	$\hat{\rho} = 0.85$	$\hat{\rho} = 0.90$	$\hat{\rho} = 0.95$	$\hat{\rho} = 0.99$
50	29.0%	27.8%	9.8%	vacuous
200	58.3%	58.3%	58.3%	24.8%
400	68.4%	68.4%	68.4%	63.9%
800	76.1%	76.1%	76.1%	76.1%
3,500	86.9%	86.9%	86.9%	86.9%

Reading

- $\hat{\Delta}_n = C(\sqrt{\log n/n} + \hat{\rho}^n)$ with $C = 2.5$
- At $n_{\text{cal}} \geq 400$: persistence $\hat{\rho}$ becomes irrelevant
- Empirical coverage (98.9%) well above theoretical floor ($\approx 88\%$)

Recommendation

- $f_c \geq 0.60$ and $T \geq 1{,}000$ for operational deployment

Conformal Intuition: A Numerical Example

Setup: 10 calibration days (illustrative $\alpha = 0.05$; paper uses $\alpha = 0.01$)

Day t	1	2	3	4	5	6	7	8	9	10
$\hat{q}_t^{\text{lo}}$	−2.1	−1.8	−2.5	−1.9	−2.3	−2.0	−2.2	−1.7	−2.4	−2.1
r_t	−1.5	−2.3	−2.0	−1.2	−3.1	−1.8	−2.8	−1.5	−2.1	−1.9
s_t^V	−0.6	0.5	−0.5	−0.7	0.8	−0.2	0.6	−0.2	−0.3	−0.2
violation?		✓			✓		✓			

Calibration

- Sorted scores: $\{-0.7,\ -0.6,\ -0.5,\ -0.3,\ -0.2,\ -0.2,\ -0.2,\ 0.5,\ 0.6,\ \mathbf{0.8}\}$
- Conformal threshold: $\lceil (n+1)(1-\alpha) \rceil$-th order statistic; here, the maximum: $\hat{q}_V^{\text{stat}} = 0.8$
- Raw rate: 3/10 violations = 30% vs 5% target; $\hat{q}_V^{\text{stat}} = 0.8$ widens VaR

Test: New day with $\hat{q}_{11}^{\text{lo}} = -2.0$

- $\widehat{\text{VaR}}_{11}^{cp} = -\left(\hat{q}_{11}^{lo} - \hat{q}_V^{\text{stat}}\right) = -(-2.0 - 0.8) = 2.8$ (raw $\widehat{\text{VaR}}_{11}^{raw} = 2.0$)
- The forecast was missing a 0.8-unit shift; conformal supplies it

ACI (Adaptive Conformal Inference) Comparison (Gibbs–Candès 2021)

Adaptive Conformal Inference

- Updates target miscoverage level α_t based on past violations
- Regret guarantee under **symmetric** miscoverage loss
- Designed for two-sided prediction intervals

Applied to VaR

- One-sided VaR structure breaks ACI's symmetric-loss assumption
- Empirical: 96.3% Green, QS 5.37×10^{-4}, width 0.040
- Wider bands than our single-split conformal at a higher Green rate

Comparison with Our Rolling Conformal

- Under stationarity: conformal shift dominates (simpler, no tuning)
- Under non-stationarity: OGD regret $O(T^{-1/2})$; drift bound $\geq 1 - \alpha - \delta_w(t) - \Delta_w$
- **Open**: unified proportional-integral-derivative (PID) regret for one-sided score + β-mixing

← back

Robustness: Calibration Fraction f_c

Design

- Vary $f_c \in \{0.50, 0.60, 0.70, 0.80\}$
  - Re-estimate $\hat{q}_V^{\text{stat}}$ and evaluate out-of-sample

Result

- Corrected $\hat{\pi}$ stable within 0.010–0.011 across all four f_c values
- Green rates similar; no material dependence on calibration fraction
- Theoretical bound tightens with n_{cal}
  - Empirical coverage already near-nominal at $f_c = 0.50$

Recommendation

- $f_c \geq 0.60$ for operational deployment
  - Below that, test-set reduction affects Kupiec power against twofold miscalibration

Robustness: Weighted Conformal

Exponentially-Weighted Nonconformity Scores

- Weighted quantile: $\hat{q}_V^{\text{stat},(\lambda)} = Q_{1-\alpha}\left(\{s_t^V\},\ \text{weights } w_t \propto e^{-\lambda(n-t)}\right)$
- With $\lambda = 0.01$ (mild decay)

Result

- Green rates and corrected $\hat{\pi}$: similar to unweighted variant
- Weighted conformal addresses non-stationarity in theory but adds no empirical benefit at our sample sizes
- The rolling variant is a stronger adaptation device

Reading

- Admissible complexity class is robust to weighting schemes

Small-Sample Behaviour

$\mathrm{Std}(\hat{q}_V^{\mathrm{stat}})$ at Varying T

- $T = 250$: $\mathrm{Std}(\hat{q}_V^{\mathrm{stat}})$ is **3–5**$\times$ larger than at $T = 5{,}000$
- $T \leq 500$ under heavy-tail misspecification: no f_c reliably achieves 75% Green
- $T \in [500, 1{,}000]$ under mild misspecification (Student-$t(5)$): adequate coverage, Green rate below 75%

Implication

- Coverage guarantees cannot compensate for extreme tail sparsity below $T \approx 1{,}000$
- For short histories (e.g. newly listed assets), consider GARCH-filtered residuals or parametric tail models

Operational Recommendation

- $T \geq 1{,}000$, $f_c \geq 0.60$

Raw Diagnostics: Before Correction (Full)

Raw violation rates ($\alpha = 0.01$, 24 assets)

Model	$\hat{\pi}$	Kupiec	Green	Red
Chronos-S	.388	24/24	0/24	24/24
Chronos-M	.419	24/24	0/24	24/24
TimesFM 2.5	.990	24/24	0/24	24/24
Moirai 1.1	.015	13/24	11/24	0/24
Moirai 2.0	.988	24/24	0/24	24/24
Lag-Llama	.029	24/24	0/24	4/24
GJR-GARCH	.004	17/24	24/24	0/24
GARCH-N	.019	16/24	8/24	0/24
Hist. Sim.	.016	13/24	12/24	0/24
EWMA	.021	19/24	5/24	0/24

Key Observations

- Chronos: $\sim 40\times$ above 1% target
- Heterogeneity spans two orders of magnitude
- All ten forecasters require some correction

Within-Family Contrast: Moirai 1.1 vs Moirai 2.0

	Moirai 1.1	**Moirai 2.0**
Predictive interface	1,000 MC samples	Nine-decile quantile grid
Raw violation rate $\hat{\pi}$	1.5%	98.8%
$\bar{R}$ / classification	0.11 / signal-preserving	3.2 / replacement

Takeaway

- Architecture and training approximately held fixed across versions
- The predictive interface emerges as a major source of the gap (reproducibility)

Cross-Sectional Correlations of $\hat{q}_V^{\text{stat}}$

Model	Ann. Vol	Kurtosis	Tail Freq.
Chronos-Small	+0.955	−0.275	+0.504
Chronos-Mini	+0.961	−0.272	+0.497
TimesFM 2.5	+0.957	−0.319	+0.528
Moirai 1.1	+0.674	−0.197	+0.524
Moirai 2.0	+0.965	−0.311	+0.516
Lag-Llama	+0.907	−0.248	+0.531
GJR-GARCH	**−0.786**	+0.245	−0.154
GARCH-N	+0.796	−0.172	+0.498
Hist. Sim.	+0.670	−0.237	+0.457
EWMA	+0.780	−0.146	+0.486

Interpretation

- **TSFMs** (six): mean volatility correlation $\bar{\rho} = 0.90$; replacement-regime (four): $\bar{\rho} = 0.96$
- **GJR-GARCH**: *negative* volatility correlation; conditional variance dynamics absorb asset-specific risk

Panel-Pooled Backtest: HAC and Cluster-Robust

Model	N_{panel}	Viol.	Corr. $\hat{\pi}$	HAC SE*	p_{Kup}	p_{cluster}
Chronos-Small	36,588	386	.0105	.0017	.295	.762
Chronos-Mini	36,588	367	.0100	.0016	.953	.986
TimesFM 2.5	36,588	423	.0116	.0018	.003	.729
Moirai 1.1	36,588	394	.0108	.0013	.144	.457
Moirai 2.0	36,588	477	.0130	.0019	.0000	.588
Lag-Llama	36,588	351	.0096	.0013	.431	.549
GJR-GARCH	38,473	442	.0115	.0010	.004	.234
GARCH-N	38,473	419	.0109	.0012	.083	.262
Hist. Sim.	38,473	415	.0108	.0018	.126	.290
EWMA	38,473	458	.0119	.0013	**.0003**	**.035**

Main Result

- ∼36,600 pooled test obs per model; all TSFMs pass cluster-robust test
- **EWMA**: only model failing cluster-robust ($p = .035$)

Known Boundary: Systemic Stress

- Cross-asset correlations spike under systemic stress
- Effective cluster count falls below 24

Multi-Quantile Stability

	$\alpha = 0.01$		$\alpha = 0.025$		$\alpha = 0.05$		$\alpha = 0.10$	
Model	$\hat{\pi}$	Rej.	$\hat{\pi}$	Rej.	$\hat{\pi}$	Rej.	$\hat{\pi}$	Rej.
Chronos-Small	.011	10/24	.024	14/24	.048	16/24	.097	15/24
Chronos-Mini	.010	13/24	.024	13/24	.047	15/24	.095	17/24
TimesFM 2.5	.013	21/24	.029	22/24	.051	22/24	.096	22/24
Moirai 1.1	.011	10/24	.025	6/24	.050	5/24	.101	5/24
Moirai 2.0	.015	20/24	.031	22/24	.052	21/24	.097	21/24
Lag-Llama	.009	2/24	.024	7/24	.049	7/24	.099	6/24

Takeaway

- **Systematic, not α-specific**: corrected $\hat{\pi}$ centres at target at all four levels
- **Moirai 1.1, Lag-Llama**: stable Kupiec performance across levels
- TimesFM 2.5, Moirai 2.0: higher rejection rates consistent with replacement regime

Diebold–Mariano Tests: Quantile-Score Comparison

	Chr-S	Chr-M	TFM	Moi-1.1	Moi-2.0	L-Llama
GJR-GARCH	.010	.005	.265	$<.001$	.052	.012
GARCH-N	.001	$<.001$	.025	$<.001$	.004	$<.001$
Hist. Sim.	.061	.139	$<.001$	.001	.058	.144
EWMA	.001	$<.001$	.042	$<.001$	.006	.004

Key Findings

- GARCH-N dominates all TSFMs; GJR-GARCH most but not TimesFM ($p = .265$)
- Lag-Llama closest TSFM (QS $= 5.2 \times 10^{-4}$)

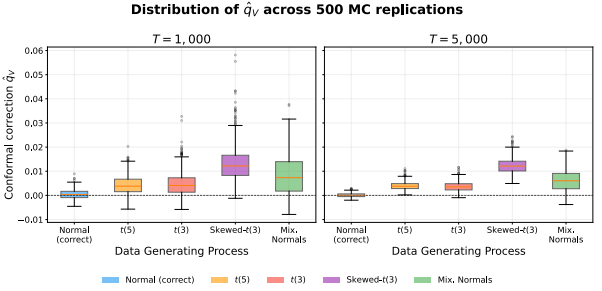
Implication

- Classical fitted models achieve lower per-asset QS
- TSFMs add value for scalable zero-shot deployment
- Conformal fixes *reliability*, not *sharpness*

Variance Specification

- Driscoll–Kraay panel HAC variance estimator; Bartlett kernel, Newey–West bandwidth

Monte Carlo: 5 DGPs (Data-Generating Processes)



DGPs: increasing misspecification

- DGP 1 (correct spec): $\hat{q}_V^{\text{stat}} \approx 0$
- DGP 2–5 (Student-t, skew-t, GARCH-t): $\hat{q}_V^{\text{stat}}$ shifts positive
- $T = 5{,}000$: Green 96–98%
- $T = 1{,}000$: Green 76–81%

Three Confirmations

- Coverage converges to nominal as $T \uparrow$
- Green rate scales as $O((\alpha T)^{-1/2})$
- $\hat{q}_V^{\text{stat}}$ increases monotonically with tail thickness/skewness

Fissler–Ziegel Joint VaR–ES (Expected Shortfall) Scores

Model	Raw FZ_0	Corrected FZ_0	Improvement %
Chronos-Small	–	−2.96	–
Chronos-Mini	–	−2.99	–
TimesFM 2.5	–	−2.30	–
Moirai 1.1	−2.45	−3.08	25.7
Moirai 2.0	–	−2.26	–
Lag-Llama	−2.49	−3.10	24.6
GJR-GARCH	−3.18	−3.25	2.1
GARCH-N	−3.17	−3.26	2.8
Hist. Sim.	−2.93	−3.06	4.6
EWMA	−3.10	−3.23	4.2

Interpretation

- Rankings consistent with QS; Lag-Llama closest TSFM
- Raw FZ_0 undefined for replacement-regime TSFMs (raw VaR meaningless)
- ES correction is heuristic; FZ scores reported as supplementary diagnostics

ES Correction: Z_2 Backtest

	Mean Z_2		Pass (x/24)	
Model	Raw	Corrected	Raw	Corrected
Chronos-Small	+131	+11	24/24	24/24
Chronos-Mini	+163	+12	24/24	24/24
TimesFM 2.5	+3.6	+3.5	24/24	24/24
Moirai 1.1	+1.6	+1.2	24/24	24/24
Moirai 2.0	+1.4	+1.1	24/24	24/24
Lag-Llama	+4.1	+1.9	24/24	24/24
GJR-GARCH	+1.6	+2.4	24/24	24/24
GARCH-N	+3.1	+2.1	24/24	24/24
Hist. Sim.	+2.9	+2.2	24/24	24/24
EWMA	+3.2	+2.2	24/24	24/24

Reading

- All pass both raw and corrected, reflecting ES **over-prediction**
- Replacement-regime: correction absorbs excess tail severity

Limitation of Z_2

- ES over-prediction inflates denominator
- Formal ES conformal guarantee remains open

Dedicated VaR Models: EVT-POT and FHS

Construction

- Both use GARCH(1,1) pre-filter; EWMA ($\lambda = 0.94$) fallback when GARCH fails
- **EVT-POT** (McNeil 2000): Generalised Pareto fit to standardised residual exceedances
- **Filtered HS** (Barone-Adesi 1999): empirical α-quantile of standardised residuals

Short-Window Performance (24 assets)

- EVT-POT: Green 18/24 (75.0%); QS 5.5; Width 0.036
- FHS: Green 13/24 (54.2%); QS 5.6; Width 0.036
- GARCH convergence failure: $\sim$70% across 24 assets $\rightarrow$ EWMA fallback does most of the work
- GPD convergence failures: $>$10% on all 24 assets

Context

- Under longer samples or tuned thresholds, EVT may perform differently

Rolling versus Static Conformal Correction

Model	*Static (70/30)*			*Rolling (250-day)*		
	$\hat{\pi}$	Green	CC pass (Cond. Cov.)	$\hat{\pi}$	Green	CC pass (Cond. Cov.)
Chronos-S	.011	19/24	4/24	.009	24/24	5/24
Chronos-M	.010	19/24	5/24	.009	24/24	5/24
TimesFM 2.5	.013	19/24	0/24	.011	22/24	3/24
Moirai 1.1	.011	21/24	24/24	.009	24/24	24/24
Moirai 2.0	.015	19/24	0/24	.011	23/24	2/24
Lag-Llama	.009	24/24	2/24	.009	24/24	2/24
GJR-GARCH	.011	19/24	7/24	.009	24/24	8/24

Key Result

- Rolling: Green rate 84.6% → **98%**; Chronos-S/M reach 24/24
- Robust to $w \in \{125, 250, 500\}$: Green 97–99%

Finite-Sample Guarantee

- Rolling windows break exchangeability; no finite-sample conformal guarantee in the classical sense
- Two operational bounds substitute: time-average $O(T^{-1/2})$ and per-window drift

← back

Panel by Asset Class: Static vs Rolling

		Static (70/30)		Rolling (w=250)	
Asset class	N	Corr. $\hat{\pi}$	Green	Corr. $\hat{\pi}$	Green
Equity	11	.010	87/99 (88%)	.009	95/99 (96%)
Bond	3	.010	23/27 (85%)	.009	27/27 (100%)
Commodity	4	.016	23/36 (64%)	.011	34/36 (94%)
Crypto	2	.005	18/18 (100%)	.010	18/18 (100%)
FX	4	.009	36/36 (100%)	.010	36/36 (100%)



Takeaway

- Rolling lifts overall Green from 84% to 97%; bonds and commodities gain most
- Commodities (64% $\rightarrow$ 94%) driven by NATGAS and WTI regime-switching volatility

$\hat{q}_V^{\text{stat}}$ Interpretation: Sign, Magnitude, Units

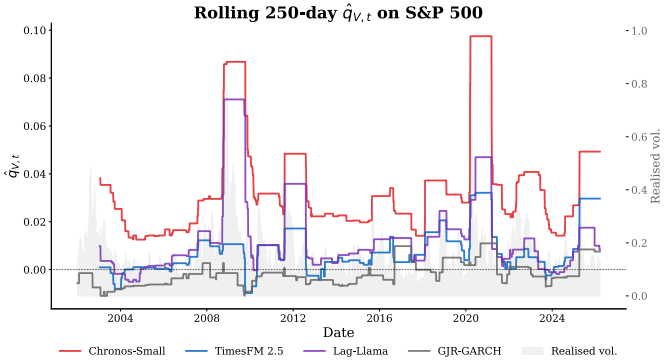
Sign Interpretation

- $\hat{q}_V^{\text{stat}} > 0$: under-covers $\rightarrow$ widen by $\hat{q}_V^{\text{stat}}$ (Chronos: $+0.038$)
- $\hat{q}_V^{\text{stat}} < 0$: over-conservative $\rightarrow$ tighten by $|\hat{q}_V^{\text{stat}}|$ (GJR-GARCH: -0.007)
- $\hat{q}_V^{\text{stat}} \approx 0$: near-calibrated (Moirai 1.1: $+0.004$, Hist. Sim.: $+0.004$)

$\hat{q}_V^{\text{stat}}$ Is Not a Test Statistic

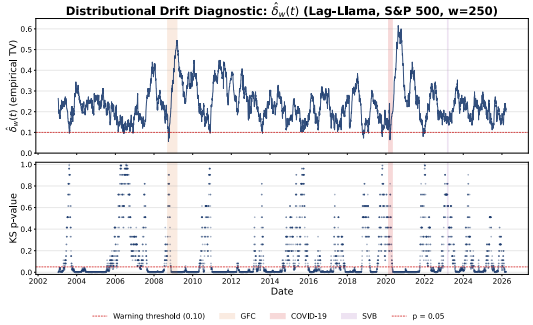
- **Physical units**: return units, directly actionable
- No p-value or critical value needed

Rolling $\hat{q}_V^{\text{roll}}$: Regime-Dependent Diagnostic



- **Construction**: rolling 250-day $\hat{q}_V^{\text{roll}}$ with realised-vol overlay
- **Behaviour**: rises in crises (2008, COVID-19, 2022), falls in calm markets
- **Use**: continuous early warning, detects miscalibration before Kupiec

Distributional Drift Diagnostic: $\hat{\hat{\delta}}_w(t)$



- **Metric**: empirical total variation (TV) distance between first/second halves of each w=250 window
- **Spikes** identify regime transitions (GFC, COVID-19, SVB) where the bound weakens
- **Operational use**: when $\hat{\hat{\delta}}_w(t) > 0.10$, shorten window or pause deployment

Regime Stability: No Permanent Drift

The Question

- Does conformal recalibration "remember" regime shifts?

Monte Carlo Evidence

- **Design**: embedded stress regime in stationary simulation
- **Behaviour**: $\hat{q}_V^{\text{roll}}$ spikes, then reverts to pre-stress level
- **Test**: pre vs post $\hat{q}_V^{\text{roll}}$ means, $p > 0.30$
- **Conclusion**: no permanent recalibration bias

Interpretation

- Trailing window is backward-looking with **finite memory**
- $\hat{q}_V^{\text{roll}}$ returns to regime-appropriate values once stress exits

Four Failure Modes

Mode 1: Small Calibration Set ($n_{\text{cal}} < 200$)

- Too few tail observations for stable $\hat{q}_V^{\text{stat}}$; Δ_n becomes vacuous

Mode 2: High Persistence ($\hat{\rho} > 0.95$)

- Mixing too slow; β-mixing remainder dominates; effective sample size collapses

Mode 3: Over-Conservative Forecaster ($\hat{q}_V^{\text{stat}} < 0$)

- Correction narrows VaR $\rightarrow$ may lose coverage under non-stationarity

Mode 4: Structural Breaks

- Shock arrives before calibration window absorbs it
- First $w/2$ days of unprecedented regime remain under-protected

Practical Safeguards

Operational Mitigants

- **Rolling 250-day** $\hat{q}_V^{\text{roll}}$: within-regime adaptivity
- **Floor** $\hat{q}_V^{\text{roll}} \geq 0$: never narrow VaR under regulation
- **GARCH pre-filter**: reduces persistence before conformal step
- **Stressed window** (Basel 2.5): conservative $\hat{q}_V^{\text{stat}}$ by including stress periods
- **Vol-triggered adjustment**: early warning at regime shifts

Bottom Line

- Not a silver bullet, but failure modes are **identifiable ex ante**
- Safeguards are cheap, interpretable, and already standard in risk desks

Practitioner Guidance: Deployment Scenarios

Scenario 1: Single-Asset Regulatory VaR

- **Recommended**: GJR-GARCH (fitted per asset) + single-split conformal, lowest width, strongest single-asset benchmark

Scenario 2: Zero-Shot Deployment Across Large Asset Universes

- **Recommended**: Moirai 1.1 or Lag-Llama + rolling conformal
  - No per-asset estimation; 97.5% Green zone across 240 model–asset pairs
  - **Caution**: screen with audit ratio $R = |\hat{q}_V^{\text{stat}}|/|\widehat{\text{VaR}}^{raw}|$ before deployment

Scenario 3: Screening Tool for Ensembles and Model Selection

- **Use** $|\hat{q}_V^{\text{stat}}|/|\widehat{\text{VaR}}^{raw}|$ **ratio** to classify forecaster quality:
  - < 1: base forecast retains predictive content (signal-preserving recalibration)
  - $\gg 1$: correction dominates (effective replacement)

Why Not Fine-Tune the TSFMs?

Zero-Shot Is the Audit Population

- The paper audits off-the-shelf TSFM checkpoints as deployed in practice
- Fine-tuning addresses a different question: can a modified model avoid the correction?

Fine-Tuning Recipes Are Not Standardised

- Tail-aware loss functions for fine-tuning TSFMs are not established
- Per-asset fine-tuning eliminates the zero-shot scalability advantage

Conformal Layer Is Forecast-Agnostic

- Fine-tuned TSFM + conformal is a natural future extension
- The conformal wrapper applies identically to any base forecaster

Computational Cost of the Rolling Implementation

Per-Asset Cost

- Rolling computation: $O(w)$ per day for a single quantile sort ($w = 250$)
- Full panel: $O(w \times N_{\text{assets}} \times N_{\text{models}} \times T_{\text{test}})$; tractable on a single core

Wall-Clock Time

- Full audit (24 assets $\times$ 10 forecasters $\times$ $\sim$6,500 days, all metrics): under one hour
- Bottleneck is TSFM inference, not the conformal layer

Baselines: Dedicated VaR Models (Detail)

Per-Asset Models (24 assets, not panel-scalable)

Method	Complexity	$\hat{\pi}$	Kup. rej.	QS ($\times 10^4$)	Width	Green %
EVT-POT	param.	.014	10/24	5.5	.036	75.0
Filtered HS	param.	.016	13/24	5.6	.036	54.2
Conformal rolling	1 shift	.010	13/240	5.63	.044	97.5

Context

- GARCH convergence failure: $\sim$70% on 24 assets $\rightarrow$ EWMA fallback does most work
- GPD convergence failures: $>$10% on all assets
- Under longer samples or tuned thresholds, EVT may perform differently

Within-Family Contrast: Moirai 1.1 vs 2.0

- Same architecture family; Moirai 1.1 ($\hat{\pi} = 1.5\%$, $\bar{R} = 0.11$) vs Moirai 2.0 ($\hat{\pi} = 98.8\%$, $\bar{R} = 3.2$)
- The **predictive interface** (MC samples vs quantile grid) emerges as a major source of the gap

Two Regimes: Full Table

	$\hat{\pi}$		QS $\times 10^4$					
Model	Raw	Corr.	Raw	Corr.	Width	W/GJR	**Green**	$\bar{R}$
Panel A: Signal-preserving ($\bar{R} < 1$)								
Moirai 1.1	.015	.011	5.4	5.3	.039	1.00	21/24	0.11
Lag-Llama	.029	.009	5.9	5.2	.041	1.05	24/24	0.36
GJR-GARCH	.004	.011	5.0	4.9	.039	1.00	19/24	0.16
GARCH-N	.019	.010	4.9	4.7	.036	0.93	21/24	0.17
Hist. Sim.	.016	.011	5.7	5.6	.039	1.00	22/24	0.11
EWMA	.021	.011	5.1	4.9	.037	0.93	20/24	0.18
Panel B: Replacement regime ($\bar{R} \gg 1$)								
Chronos-S	.388	.011	37.9	5.8	.040	1.03	19/24	17.3
Chronos-M	.419	.010	41.1	5.7	.041	1.04	19/24	23.5
TimesFM 2.5	.990	.013	334	9.5	.069	1.75	19/24	3.2
Moirai 2.0	.988	.015	305	9.1	.063	1.60	19/24	3.2

Headline

- Panel A: 127/144 Green (88.2%); CC pass 86.1%
- Panel B: 76/96 Green (79.2%); CC pass 40.6% (empirical-quantile fallback)

Three Strands of Literature

Time Series Foundation Models (TSFMs) for Financial Data

- Chronos, TimesFM 2.5, Moirai 1.1/2.0, Lag-Llama (2024): zero-shot, no per-asset fitting
- Miscalibration evidence: Goel (2024), Pele et al. (2026)

Regulatory Backtesting

- Kupiec (1995), Christoffersen (1998), Basel (1996): diagnostic, not prescriptive

Model-Free Post-Processing

- Conformal: Vovk (2005); Romano (2019) Conformalized Quantile Regression (CQR)
- Dependence-aware: Barber (2023), Gibbs–Candès (2021), two-sided, full-distribution

Why Is This Surprising?

TSFMs Succeed in the Bulk, Fail in the Tail

- Strong in the bulk (mean forecasting, density estimation)
- Tail is where risk lives, and where they fail
- Good overall accuracy $\neq$ good tail calibration

One Plausible Mechanism (Observational, Not Causal)

- Pre-training corpus under-represents fat tails
  - Mostly non-financial (weather, demand, traffic); prior under-weights extremes
- Consistent with $\mathrm{Corr}(\hat{q}_V^{\mathrm{stat}}, \mathrm{vol})$: $+0.96$ replacement-regime TSFMs vs. -0.79 GJR-GARCH
- **Causal mechanism cannot be identified**: observational evidence only

Implication

- Mean-accuracy benchmarks not informative for tail risk
- Tail calibration must be evaluated and corrected separately

The Reframing: Complexity Selection, Not Method Selection

Our Reframing: Complexity Selection, Not Method Selection

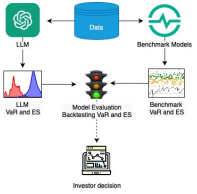
- A single scalar $\hat{q}_V^{\text{stat}}$ is the *operational complexity ceiling*, constrained by tail-sample size
- Ex ante, DGP-independent: any M-estimator on the tail sub-sample inherits the bound
- The question is not *which* correction, but *how many parameters* the tail can support

Context

- At $\alpha = 0.01$, $T \approx 1{,}500$: only ~ 15 tail observations
- $k^* = \arg\min_k \{\mathcal{B}(k)^2 + \mathcal{V}(k)\}$; $\mathcal{V}(k) \sim k/(\alpha T)$ dominates at $\alpha T \approx 15$

LLM (Large Language Model) VaR and LLM-ES (ESWA, 2026)





Key Finding of the ESWA Paper

- LLM-VaR/ES perform comparably to classical models (3 GPT models, 9 assets)
  - GPT-4/4o show systematic tail miscalibration $\rightarrow$ motivates post-hoc recalibration
- **This paper**: 6 TSFMs + 4 classical benchmarks $\times$ 24 assets

Notation (Symbols Not Defined in Main Slides)

Theory

- f_S: density of the score S_t at the target quantile
- g_n: gap (unused observations) between calibration and test
- $\beta(k)$: β-mixing coefficient at lag k
- $\Delta_n \leq C\sqrt{\log n/n + \beta(g_n)}$: coverage gap bound
- $q_{1-\alpha}$: population $(1-\alpha)$-quantile of S_t

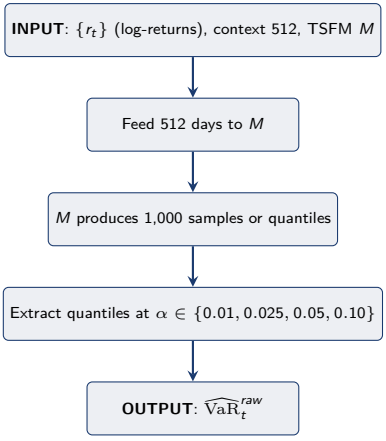
Sample Structure

- n_{cal}: calibration set size
- $w = 250$: rolling window width (trading days)

Basel Regulation

- m_c: Basel capital multiplier (3.00 Green, 3.40–3.85 Yellow, 4.00 Red)

TSFM-VaR Pipeline



Benchmarks: Same Interface

- GJR-GARCH, GARCH-N, Hist. Sim., EWMA: 250-day rolling fit
- Produce a distribution; extract α-quantile
- Same conformal layer applied downstream

Experimental Design

Two Windows, Two Roles

- **TSFM context**: 512 daily returns; fixed input, no parameters re-estimated
- **Parametric window**: 250 obs (Basel); GJR/GARCH-N/EWMA/HS re-fit each step
- No fine-tuning; all checkpoints predate evaluation period

Calibration Split

- **Static**: 70% / 30% (chronological) $\rightarrow \approx$ 1,500 test obs per asset
  - Pooled: $\approx$ 36,600 test obs per TSFM
- **Rolling**: 250-day trailing window, re-estimated daily

Reproducibility

- HuggingFace checkpoints pinned; Yahoo Finance data
- Full pipeline in Quantlet repo

VaR Backtests: Coverage and Independence

Kupiec (1995): Unconditional Coverage

- Likelihood Ratio (LR) test, Proportion of Failures (POF):

$$\mathrm{LR}_{\mathrm{POF}} = -2\ln\frac{(1-\alpha)^{T-x}\alpha^{x}}{(1-\frac{x}{T})^{T-x}(\frac{x}{T})^{x}} \sim \chi^2_1$$

- x = number of VaR violations, T = sample size, $\hat{\pi} = x/T$
- Tests $H_0 : \hat{\pi} = \alpha$; ignores clustering

Christoffersen (1998): Conditional Coverage

- Adds independence test via first-order Markov transition matrix

Basel Traffic Light (BCBS 1996)

- Count-based rule over 250 trading days:
  - Green: ≤ 4 violations $\rightarrow m_c = 3.00$
  - Yellow: 5–9 violations $\rightarrow m_c \in [3.40, 3.85]$
  - Red: ≥ 10 violations $\rightarrow$ model disallowed

ES Backtest and the Diagnostic Gap

Acerbi Z_2 (2014): ES Magnitude Test

- Tests whether realised losses on violation days match the predicted ES:

$$Z_2 = \sum_t \frac{V_t \, r_t}{T \alpha \, \widehat{\mathrm{ES}}_t^{\alpha}} + 1$$

- $V_t = \mathbf{1}\{r_t < -\widehat{\mathrm{VaR}}_t^{\alpha}\}$: violation indicator; negative Z_2: ES under-predicts severity
- Complements coverage tests by checking **magnitude**, not just frequency

The Diagnostic Gap

- All three tests (Kupiec, Christoffersen, Acerbi) **detect** miscalibration
  - They do **not prescribe** a correction
- $\hat{q}_V$ fills this gap: continuous, signed, directly actionable

← back

Backup: Alternative Recalibration Methods

Post-Hoc Methods (applied to forecaster output)

- **Scale** ($k = 1$, mult.): $\widehat{\mathrm{VaR}}_t^{\,scale} = c\,\widehat{\mathrm{VaR}}_t^{\,raw}$
- **Historical quantile** ($k = 0$): empirical α-quantile of calibration returns
- **QR on residuals** ($k = 2$): linear QR of r_t on $\hat{q}_t^{lo}$
- **ACI** (Gibbs–Candès 2021, $k = 1$ seq.): online update of α_t from past violations
- **Isotonic** ($O(n)$): monotone non-parametric mapping of violation indicators
- **GBM-QR**: LightGBM with pinball loss on $\hat{q}_t^{lo}$ + lagged volatility
- **GAMLSS-SST** ($k = 7$, Rigby–Stasinopoulos 2005): Skew Student-t with covariate dependencies on μ, σ, ν, τ

Dedicated VaR Models (per-asset, not post-hoc)

- **EVT-POT** (McNeil 2000): Generalised Pareto fit to standardised GARCH residual exceedances
- **Filtered Historical Simulation** (Barone-Adesi 1999): empirical α-quantile of GARCH-standardised residuals

Backup: Regime Classification Sensitivity

Assets (out of 24) classified as replacement regime

	$R > 1.2$			$R > 1.5$			$R > 2.0$		
	10 d	20 d	50 d	10 d	20 d	50 d	10 d	20 d	50 d
Panel A: Signal-preserving									
Moirai 1.1	8	4	2	5	2	0	3	2	0
Lag-Llama	8	3	1	5	2	0	2	0	0
GJR-GARCH	0	0	0	0	0	0	0	0	0
GARCH-N	8	5	1	2	1	0	1	0	0
Hist. Sim.	4	3	2	2	2	1	1	1	1
EWMA	**14**	**14**	11	**13**	12	5	4	1	1
Panel B: Replacement									
Chronos-Small	24	24	24	24	24	24	24	24	24
Chronos-Mini	24	24	24	24	24	24	24	24	24
TimesFM 2.5	24	24	24	24	24	24	24	24	24
Moirai 2.0	24	24	24	24	24	24	24	24	24

Takeaway

- Panel B: 24/24 under all 9 combinations; EWMA only boundary case
- Classification is data-driven, not threshold-sensitive

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Replacement Regime: Operational Definition

Replacement Ratio

$$R = \frac{|\hat{q}_V^{\text{stat}}|}{|\widehat{\text{VaR}}^{raw}|} \quad \text{(correction magnitude relative to raw VaR)}$$

Two Regimes

- $\bar{R} < 1$: signal-preserving; conformal applies a minor shift, base-model content preserved
- $\bar{R} \gg 1$: effective replacement; conformal dominates the raw forecast, coverage via empirical fallback
- Operational rule: $R_t > 1$ persistent for ≥ 20 trading days triggers replacement classification

Order-of-Magnitude Gap

- No forecaster has stable $\bar{R} \in (0.7, 1.8)$
- Classification robust to threshold choice (sensitivity table)

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Legacy LLM Benchmark (Pele et al., 2026)

GPT-3.5/4/4o Results (9 assets, 6 windows)

Model	$\overline{\hat{q}_V^{\text{stat}}}$	Green	Z_2 Pass	Assets
GPT-3.5	$+0.002$	8/9	9/9	9
GPT-4	$+0.024$	9/9	9/9	9
GPT-4o	$+0.020$	9/9	9/9	9

Key Points

- 144/162 Green overall (108/108 GPT-4/4o)
- Newer LLMs incompatible with LLMTime protocol $\rightarrow$ TSFMs used here
- TSFMs: purpose-built, open-weight, reproducible

Outlook: Fine-Tuning, FRTB, and Theory

(1) Fine-tuning TSFMs on Financial Data

- **Conjecture**: fine-tuning reduces $|\hat{q}_V^{\text{stat}}|$ but does not eliminate need for CP

(2) FRTB (Fundamental Review of the Trading Book): VaR → ES

- **Three candidate pathways**
  - Conformal Risk Control (Angelopoulos 2024)
  - Joint Fissler–Ziegel nonconformity score; EVT tail-shape properification
- Unified distribution-free ES solution: open

(3) Theory (partially resolved)

- **Solved**: OGD regret bound (dependence-free $O(T^{-1/2})$ time-average miscoverage)
- **Solved**: non-exchangeable drift bound with measurable $\delta_w(t)$
- **Open**: unified PID regret combining one-sided score + β-mixing

(4) Software

- Python package `conformal-oracle` available on PyPI: `pip install conformal-oracle`

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